

Partner-Specific Affective Precision in Social Active Inference

Anonymous Author(s)

Anonymous Institution

Abstract. Affect—the fundamental experiential substrate of emotion in psychology and neuroscience—does not merely colour inner experience. It regulates how decisively beliefs become action, mediating the transition from what an agent knows to what it commits to. Active inference formalises this as a metacognitive signal: a running estimate of generative model fitness that modulates action confidence through inferred policy precision [12]. But this account has a structural gap at the multi-agent level. Social confidence is relational. We carry different degrees of confidence toward different partners, learned from each partner’s behavioural history. We therefore extend affective precision to the partner level: a focal agent maintains a separate model-fitness estimate β_k for each relationship, updated from that partner’s action evidence and deployed as a partner-specific action-confidence budget.

Simulations in repeated multi-partner trust games show that this mechanism is informative across several social decision regimes. In open graded choices, partner-local affective precision reliably changes policy deployment: policy entropy falls under the precision-coupled condition relative to no-affect and tracked-only controls, while payoff advantages remain task-dependent. A tracked-only lesion, which updates β_k but prevents it from setting policy precision, localises the behavioural effect to the action-confidence pathway rather than to improved partner-belief quality. A reliability-versus-reward diagnostic clarifies the central boundary of the present model: the architecture is designed to track predictive reliability, but the current task does not yet cleanly establish a reward-independent model-fitness readout. A local-versus-global precision comparison shows that partner-local β_k preserves cleaner partner-specific signal quality, whereas aggregate payoff can remain similar when the environment does not strongly punish cross-partner averaging. Under abrupt betrayal, the corrected partner-choice selector produces the strongest behavioural anchor: partner-local precision improves payoff while lowering joint state accuracy, demonstrating a metacognitive rather than epistemic route to adaptive social action.

We then use the same mechanism as a scaffold for individual-difference modelling. Varying precision gain α and the prior over model fitness produces psychologically interpretable trust-calibration profiles—anxious-reactive, hypervigilant, naive-stubborn, and avoidant-rigid—whose predicted signatures span exploitation sensitivity, betrayal recovery, forgiveness, and mixed-volatility discrimination. Affect, in this account, is a relationship-specific metacognitive regulator of social commitment. Its

value lies not in making agents universally more rewarding, but in explaining when beliefs become action, when confidence helps or harms, and how variation in confidence dynamics can produce recognisable forms of social trust.

Keywords: Active inference · Affect · Social cognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social life requires two things that are routinely conflated. The first is *knowing*: learning what partners are likely to do, building representations of their dispositions and intentions. The second is *committing*: deciding how decisively to act on what you know. A well-calibrated belief that a partner defects 80 % of the time is not the same as knowing how much to bet on it. The inference problem and the commitment problem are separable, and most formal models of social cognition address the first while leaving the second implicit.

In psychology and neuroscience, affect—the fundamental experiential substrate of emotion [2], the underlying valence and arousal that gives emotional experience its character—is what regulates the commitment problem. Affect does not merely colour experience; it mediates the transition from inference to action, modulating how much weight beliefs carry when choices are made. This is computationally significant, not merely phenomenological: an agent that infers well but commits poorly wastes its social knowledge, while one that commits confidently on fragile beliefs exposes itself to systematic exploitation. The affective signal, in this view, is not about representing the social world—it is about regulating how strongly the agent’s representation of the social world drives its behaviour.

The active inference framework [7, 8, 15] provides a formal route from this observation to a computational mechanism. Agents minimise variational free energy to update beliefs and evaluate candidate policies by their expected free energy [9, 6]. Action selection is precision-weighted: the same beliefs can produce decisive or tentative behaviour depending on γ , which controls how sharply the agent commits to its most-preferred policy. Hesp et al. [12] made affect endogenous by modelling the agent as actively inferring a higher-order belief β over its expected action precision—the subjective fitness of its current generative model. An affective charge signal ϕ drives updates to β : it rises when the model performs better than a prior baseline and falls when it performs worse. Affect, in this formalisation, is a derivative of model fitness rather than reward history—a distinction that allows the signal to be positive toward a reliably hostile partner (whose actions the model predicts well) while remaining informative about when to act confidently.

This formalisation has a gap at the multi-agent level. Emotional confidence is inherently relational. In daily social life, we do not carry the same felt certainty toward every person we know—we carry it relationship by relationship, earned

through each partner’s specific behavioural history. A person can have a well-validated model of a longtime colleague, a fragile model of a new acquaintance, and an actively revising model of a partner who has recently surprised them, and these three states of social confidence should not share a single global precision budget. Global precision conflates them: it cannot express that you should act decisively with a well-understood partner and cautiously with an unfamiliar one, even when both beliefs are held simultaneously. The psychology of affect demands localisation; the only question is what such localisation produces.

We extend the affective precision mechanism to the partner level. A focal agent interacting with multiple partners maintains a separate belief state and a separate β_k estimate for each partner k , both updated from that partner’s behavioural evidence alone. This allows action confidence to be modulated independently across the agent’s social portfolio—more decisive toward partners whose behaviour the model predicts well, more cautious toward those who remain uncertain. The interest of this extension lies not in the factorisation itself, which is forced by the relational structure of affect, but in the dynamics it produces and what those dynamics reveal about the computational basis of social behaviour.

Using a repeated multi-partner trust game with varied decision regimes, we show that the emergent behavioural dynamics of partner-local affective precision are psychologically recognisable. The mechanism changes how partner beliefs are deployed, not merely what beliefs are learned: in open graded regimes it lowers policy entropy, and in lesion controls this effect disappears when β_k is tracked but decoupled from policy precision. Local precision also separates signal quality from behavioural gain. It preserves a cleaner partner-specific relationship between precision and prediction error than a global tracker, but global precision can still perform similarly when the environment does not strongly punish cross-partner averaging. Under abrupt betrayal, partner-local precision becomes especially informative: it can improve payoff while lowering joint state accuracy, showing that the behavioural effect is metacognitive rather than epistemic. The same mechanism also has a boundary condition. A reliability-versus-reward diagnostic shows that the current task does not yet cleanly establish that model-fitness precision is reward-independent in behaviour, even though the architecture is built from partner-action surprisal. This unresolved diagnostic is part of the contribution because it identifies exactly what a publication-ready model-fitness test must separate.

Going further, systematic manipulation of the precision-gain parameter α and the prior over model fitness produces four behavioural phenotypes—anxious-reactive, hypervigilant, naive-stubborn, and avoidant-rigid—that differ in exploitation sensitivity, betrayal recovery, forgiveness speed, and mixed-volatility discrimination. These signatures map onto empirically characterised variation in human personality and clinical profiles of social trust, connecting the computational architecture to individual differences. The phenotype predictions are consistent with the empirical literature on individual differences in social learn-

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	-1
Agent defects	5	1

ing rates [3] and volatility tracking [14], and with computational accounts of trust dysregulation in affective and personality disorders [1].

We organise the paper around three claims rather than around hypothesis labels. First, partner-local affective precision is a deployment mechanism: it changes the confidence with which partner beliefs become action. Second, its usefulness depends on social temporal structure: stable relationships, cross-partner heterogeneity, and abrupt betrayal each expose different benefits and liabilities of confidence accumulation. Third, variation in the mechanism’s gain and priors produces a family of trust-calibration phenotypes that can be linked to human individual differences. The resulting account is deliberately not a claim that affect always improves reward. It is a claim that affective precision supplies the missing metacognitive layer between social inference and social commitment.

2 Methods

2.1 Trust game

A focal active-inference agent plays a repeated trust game [4, 13] with four partners over an episode of up to 200 rounds. Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *disposition* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent’s payoffs. The structure creates a genuine inference problem: defecting is individually tempting regardless of partner behavior, but a cooperative partner is worth engaging, so the agent must infer the partner’s type and disposition to choose well. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. This distributes policy probability mass across several options, giving partner-local affective precision meaningful room to influence which level is selected. A partner-choice regime adds a prior decision each round: the agent selects which partner to engage before choosing an action, turning approach and avoidance into explicit policy outputs.

2.2 Per-partner generative models

The focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k , rather than a single pooled social model. This keeps evidence about

each partner’s behavior independent and—critical for the analysis below—allows the confidence with which each model is acted upon to differ across partners. For partner k :

$$s_k = (s_k^{\text{type}}, s_k^{\text{disp}}) \quad (\text{hidden: partner type and disposition}) \quad (1)$$

$$o_k = (o_k^{\text{act}}, o_k^{\text{pay}}) \quad (\text{observed: partner response and payoff}) \quad (2)$$

$$a \in \mathcal{A} \quad (\text{cooperate/defect or investment level}) \quad (3)$$

The model $\mathcal{M}_k = \{A_k^{\text{act}}, A_k^{\text{pay}}, B_k, C_k, D_k, E_k\}$ encodes likelihoods ($A_k^{\text{act}}, A_k^{\text{pay}}$), transition dynamics (B_k), prior beliefs (D_k), preferred observations (C_k), and policy priors (E_k). Payoff enters as an observation modality rather than an external reward cache, so the agent’s preferences are part of the same inference problem as its partner beliefs. State and policy inference use the official `pymdp` discrete active-inference implementation [11, 6].

2.3 Partner-local affective precision

Hesp et al. [12] treat the policy precision parameter γ as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over its expected action precision, representing subjective model fitness—how well the generative model is supporting confident action. An affective charge signal ϕ drives updates to β : it rises when the model performs better than a prior baseline and falls when it performs worse, making it a derivative of model fitness that responds to changes in predictive performance rather than to absolute reward level.

We extend this to the per-partner case. After interacting with partner k , the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q(s_k^{\text{type}}, s_k^{\text{disp}})). \quad (4)$$

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-disposition model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. We set the neutral fitness baseline at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge.

Affective charge is:

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \quad (5)$$

where σ_0^2 is a prior fitness baseline and α is a gain parameter. Positive ϕ_k means partner k ’s model predicted better than the baseline; negative ϕ_k means it predicted worse. A categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$ is updated each round by ϕ_k . Higher β_k represents lower expected model fitness—the agent has accumulated evidence that its model of this partner is unreliable. The partner-specific policy precision is:

$$\gamma_k = \gamma_{\text{base}} / \mathbb{E}_{q(\beta_k)}[\beta_k]. \quad (6)$$

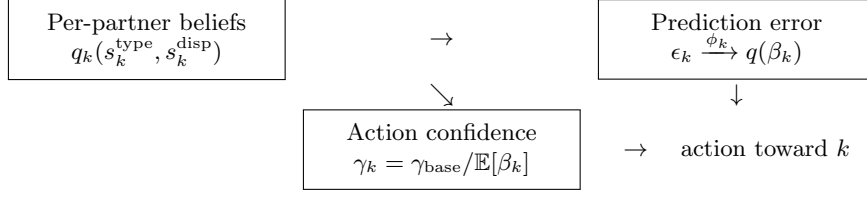


Fig. 1: Partner-local affective precision. Each partner has an independent belief state and precision estimate. Prediction error on partner action updates β_k , which sets how sharply the agent commits to policies involving that partner.

Note the direction: β_k represents precision *volatility*, not precision itself. Higher β_k means the model has been less reliable, producing a broader, less committed policy distribution. This runs opposite to the intuition that higher β equals more confidence — it is the inverse that drives action sharpening. A well-validated partner model drives $\mathbb{E}[\beta_k]$ toward low values, raising γ_k and sharpening policy selection for that partner. An unreliable partner model keeps $\mathbb{E}[\beta_k]$ high, lowering γ_k and leaving policy selection diffuse. The mechanism thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent’s ability to form prior expectations over its own future affective states (discussed in Section 4).

Tracked-only condition. To establish that any behavioral effect of the signal runs specifically through action confidence—rather than through indirect effects on belief quality—we include a tracked-only variant. This variant retains the β_k update in full but fixes $\gamma_k = \gamma_{\text{base}}$ for all partners at all times, severing the link from the precision estimate to the softmax temperature. The β_k posterior evolves normally and is available for analysis, but it never changes how policies are selected. If tracked-only behavior matches the no-affect baseline while full precision modulation differs from both, the behavioral effect runs through the γ_k pathway. If tracked-only matches full precision modulation, the effect is mediated by belief quality, not action sharpening.

3 Results

We organise the simulations by claim rather than by internal hypothesis labels. The first family of experiments establishes the mechanism: whether partner-local affective precision changes deployment, whether the signal can be separated from reward, whether locality matters, and when confidence accumulation helps or harms. The second family treats the same mechanism as a model of individual differences by manipulating the precision-gain parameter α and the prior over

model fitness. Current evidence comes from the server-side post-fix log-surprisal runs and the ongoing phenotype runs. Full per-round outputs remain in the server results tree; this manuscript uses only compact summaries and figures.

3.1 Open graded choice: affective precision changes deployment

The first requirement for the mechanism is simple: partner-local β_k must actually change policy deployment. This is only visible in regimes where the policy posterior is not already saturated. In the open graded trust game, the agent chooses among multiple investment levels and among partners, so precision can sharpen or diffuse a genuinely graded policy distribution. In this regime, the precision-coupled agent has lower policy entropy than the no-affect baseline: mean entropy is 8.59 under partner-local precision, compared with 8.79 under no affect. A global beta tracker produces a similar but slightly weaker sharpening (8.64). The same pattern appears in the deployment lesion comparison: tracked-only and no-affect controls match on the behavioural surface because both keep policy precision fixed, while the precision-coupled condition changes the policy posterior.

The payoff result is deliberately more restrained. In the open graded regime, partner-local precision does not yet improve total payoff: local precision averages 1851.3, global beta 1851.3, and no affect 1864.2. This matters because it prevents a generic “affect improves reward” reading. The immediate effect of β_k is not reward maximisation; it is action sharpening. Whether that sharpening improves reward depends on whether the environment rewards confident commitment, cautious exploration, or rapid reallocation.

The tracked-only lesion is the critical control. It retains the β_k update but removes the $\beta_k \rightarrow \gamma_k$ link. When tracked-only behaviour matches no-affect behaviour, the model rules out an interpretation in which the beta tracker secretly improves partner beliefs. The behavioural effect, when it appears, comes from the commitment pathway: the same partner model is acted on with a different degree of policy confidence.

3.2 Reliability versus reward: the diagnostic target is not yet clean

The theory predicts that β_k should track model reliability rather than partner value. The update is based on partner-action surprisal, not payoff, so a reliably hostile partner should eventually become predictable even if it remains undesirable. This distinction is central to the account because it separates affective confidence from ordinary reward learning.

The current reliability-versus-reward task is informative precisely because it does *not* yet cleanly support the strongest version of that claim. In the post-fix smoke, partner-local precision has $|\text{corr}(\text{precision}, \text{surprise})| = 0.226$ and $|\text{corr}(\text{precision}, \text{payoff})| = 0.615$ across partner-seed units. Global beta is weak on both dimensions (0.115 versus 0.103). The payoff comparison also moves against the affect condition in this task: local precision averages 492.7 versus 552.0 for no affect.

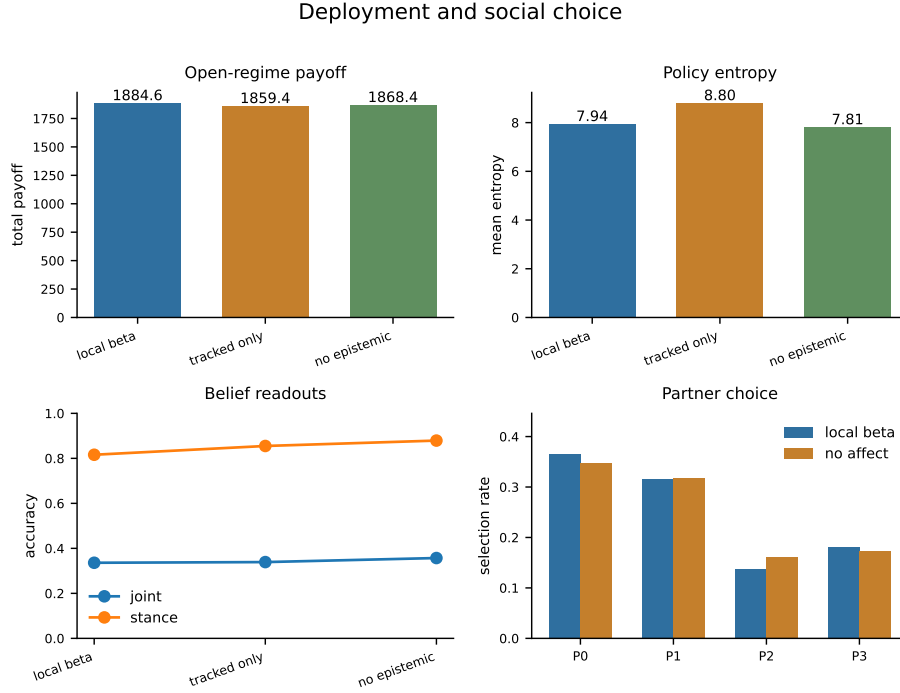


Fig. 2: Deployment and social choice diagnostics. The open graded regime shows that precision coupling changes policy entropy and deployment while payoff remains close to no-affect controls. The partner-choice panel summarises allocation across partners and motivates the later mixed-volatility setting, where allocation is the central behavioural readout rather than an auxiliary statistic.

This result should not be hidden. It tells us that the current task lets exposure, payoff, and predictability covary too strongly. The correct manuscript claim is therefore conditional: the *architecture* is a reliability tracker by construction, but the present behavioural diagnostic has not yet established a reward-independent readout. The follow-up experiment should match expected reward while varying action predictability, or match predictability while varying payoff, so that β_k can be tested against the distinction the theory actually makes.

3.3 Partner-local versus global precision: locality improves signal quality before it is behaviourally necessary

The local-versus-global comparison asks whether the relationship-specific implementation matters, or whether a single affective precision estimate over the whole social environment is enough. The focal-switch task provides a useful decomposition. Local beta produces the cleanest partner-specific diagnostic: $|\text{corr}(\text{precision}, \text{surprise})| = 0.943$ versus $|\text{corr}(\text{precision}, \text{payoff})| = 0.110$. Global

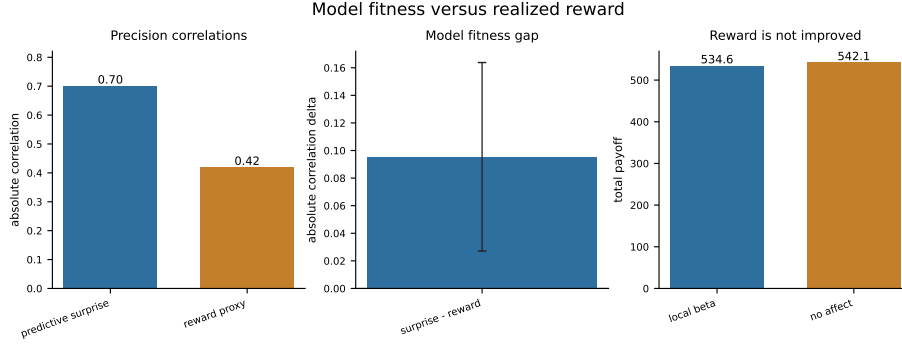


Fig. 3: Reliability-versus-reward diagnostic. The architecture computes affective charge from partner-action surprisal, but the current task does not yet isolate predictability from payoff exposure. This negative diagnostic is retained because it defines the next necessary experiment: matched-reward reliability manipulations rather than more seeds on a confounded design.

beta is much weaker on the same readout (0.149 versus 0.043), as expected if evidence from different partners is being averaged into one shared confidence state.

Aggregate payoff, however, does not yet make locality behaviourally necessary. Global beta averages 976.2, local beta 946.8, and no-affect/tracked-only controls 950.7. This is not a failure of the theory; it is a boundary condition. The focal-switch environment is sufficient to show that local beta is a cleaner relationship-specific signal, but not strong enough to force a reward advantage from that cleaner signal. If the social environment does not punish cross-partner averaging, a global confidence estimate can look behaviourally adequate despite being mechanistically less interpretable.

The partner-indexed post-switch diagnostics point in the same direction. Local beta preserves separate precision trajectories for switched and untouched partners, while global beta compresses them into one shared state. But the current window is short and aggregate payoff remains close. The stronger test is the mixed-volatility experiment in Section 3.7, where stable, exploitative, abrupt-switch, and gradual-drift partners coexist. That environment makes a clear behavioural prediction: global precision should either over-suppress confidence toward stable partners or under-suppress it toward volatile ones, while local precision should discriminate between relationships.

3.4 Abrupt betrayal: confidence helps by changing action, not by improving belief

The abrupt betrayal experiment is the strongest current behavioural anchor. One partner begins cooperative and then switches to hostile behaviour at a fixed round. Under the corrected agent-choice selector, partner-local affective precision

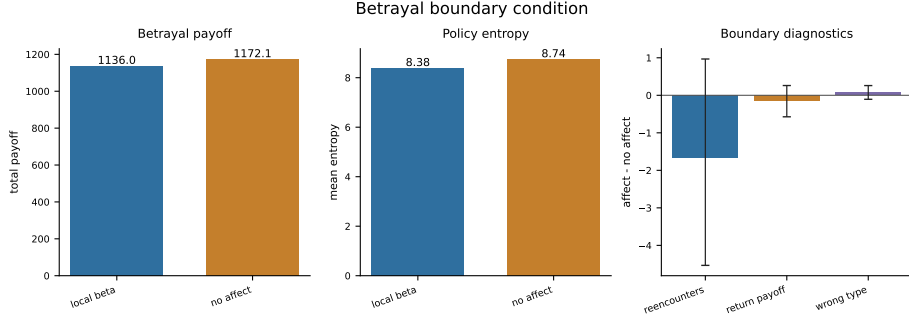


Fig.4: Abrupt betrayal boundary condition. Under the corrected selector, partner-local precision improves payoff and lowers policy entropy relative to no-affect and tracked-only controls. The same run shows lower joint accuracy, which is crucial: the gain is not better partner-state inference, but a different way of deploying action confidence after social change.

improves total payoff relative to no-affect and tracked-only controls: local precision averages 1322.3, while no affect and tracked-only average 1225.0. Global beta averages 1216.2. The precision-coupled agent also has lower policy entropy (7.47 versus 8.68 for no affect/tracked-only), showing that the behavioural difference is carried by sharper policy selection.

The accuracy pattern is the key result. Local precision improves payoff while reducing joint type-by-stance accuracy (0.319 versus 0.425 for no-affect/tracked-only). This dissociation rules out a simple epistemic explanation. The affective agent is not winning because it has a better hidden-state estimate. It is winning because confidence changes how it commits and reallocates in a partner-choice setting.

The reallocation readout shows the selectivity of the effect. After the switch, local precision reencounters the betraying partner at a lower rate than the no-affect and tracked-only baselines. The mean reencounter selection rate is 0.052 for local precision, compared with 0.322 for no-affect/tracked-only and 0.241 for global beta. This is exactly the kind of behaviour a relationship-specific confidence mechanism should support: the agent can reduce commitment toward one partner without globally lowering confidence toward every other relationship. At the same time, the result exposes the temporal risk of affective precision. Confidence accumulated during the cooperative phase can become stale after betrayal; without a separate volatility detector, the mechanism must relearn that the relationship has changed.

3.5 Social allocation: precision reorganises engagement before reward is stable

The partner-choice experiment isolates allocation as the behaviour of interest. In this setting the agent must choose whom to engage before choosing how much to

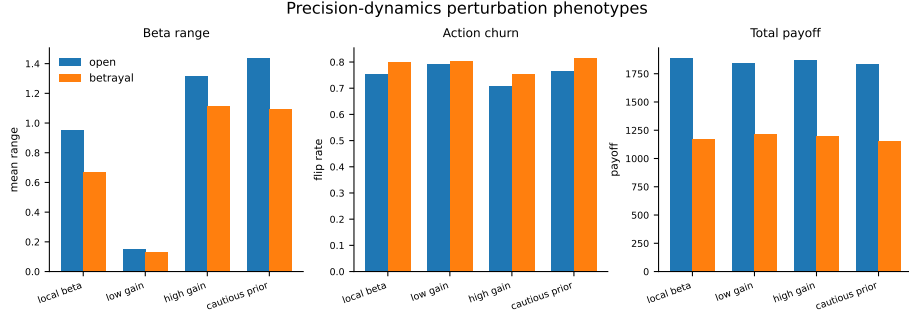


Fig. 5: Precision-dynamics perturbations. Parameter changes alter beta range, autocorrelation, entropy, and action switching without requiring corresponding improvements in payoff or partner-state accuracy. These conditions support the computational claim that social-confidence dynamics can vary independently of belief correctness; they should not be read as clinical validation.

invest. The current post-fix readout is underpowered for payoff: local precision averages 377.3, global beta 388.7, and no affect 385.3. But entropy and partner-selection patterns are still informative. Local precision slightly lowers policy entropy (4.66 versus 4.81 for no affect), indicating that the action-confidence pathway is active even when reward remains noisy.

This experiment deserves a role in the paper because partner choice is the natural behavioural surface for relational affect. Social confidence is not only about how much to invest once a partner is selected; it is also about approach, avoidance, and the distribution of attention across a social portfolio. The current partner-choice result does not support a payoff claim, but it motivates the richer mixed-volatility design: allocation should be tested where the agent has a stable safe partner, a stable bad partner, and partners whose behaviour changes over time.

3.6 Precision perturbations: beta dynamics separate from clinical claims

The perturbation experiment varies the dynamics of β_k directly. Its purpose is not to validate clinical categories, but to check that the model can produce distinct social-confidence trajectories when gain and persistence are changed. This works cleanly. The alexithymia-like low-gain condition shows strongly attenuated beta movement, with mean beta range 0.180. The default affect condition has range 1.126, the borderline-like high-gain condition 1.412, and the depression-like cautious condition 1.464.

The important point is dissociation. Payoff and accuracy are modest and not the main readout. The perturbations change the temporal profile of confidence: how far beta moves, how persistent the movement is, and how often actions flip. That makes this experiment a bridge between the mechanism results above and

the individual-difference experiments below. It shows that the model has controllable precision dynamics before we interpret those dynamics as psychologically recognisable trust-calibration profiles.

3.7 Affective precision dynamics as individual differences in social trust calibration

The second result family treats α and the prior over model fitness as individual-difference parameters. The goal is not to claim that four hand-named simulated agents exhaust human social variation. The goal is to show that small, interpretable changes to the affective precision mechanism generate distinct profiles of exploitation sensitivity, recovery, forgiveness, and discrimination in heterogeneous social environments.

The precision-gain axis Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ quantifies how reactive social confidence should be. Low- α agents should move β_k slowly: early exploitation persists longer, betrayal recovery is delayed, selection Gini remains lower, and beta range is compressed. High- α agents should move quickly: they should detect behavioural changes faster and concentrate engagement more strongly, but they should also overreact to noise and carry higher variance across seeds. The final run reports early exploitation rate [*Exp A: exploitation rate by alpha*], betrayal recovery time [*Exp A: recovery rounds by alpha*], selection Gini [*Exp A: selection Gini by alpha*], entropy trajectories [*Exp A: entropy trajectories*], and beta range [*Exp A: beta range by alpha*].

Payoff is expected to be non-monotonic. High gain can help when change detection is valuable, but hurt when local noise masquerades as social volatility. Low gain can be stable in stationary environments, but sluggish after betrayal. The key result is therefore not “which alpha wins”, but the tradeoff surface across open graded and betrayal regimes.

Phenotype quadrants: prior $\times$ gain Crossing prior belief (naive versus cautious) with α (low versus high) produces four behavioural phenotypes. The **anxious-reactive** profile (high α , naive prior) is predicted to show fast approach and fast withdrawal: high initial engagement, rapid confidence revision after negative evidence, and high variance in post-betrayal recovery. The **hypervigilant** profile (high α , cautious prior) should detect negative shifts quickly but extend trust slowly, producing a threat-sensitive asymmetry. The **naive-stubborn** profile (low α , naive prior) should remain exploitable because confidence starts high and updates slowly. The **avoidant-rigid** profile (low α , cautious prior) should under-engage even with reliably cooperative partners because positive evidence does not accumulate into decisive action.

The factorial experiment reports the full metric set for each profile: exploitation sensitivity, betrayal recovery, selection concentration, entropy trajectory, beta range, and trust asymmetry [*Exp B: phenotype metrics and radar summary*]. These phenotypes connect the model to empirical individual differences

not as labels imposed after the fact, but as predicted signatures of gain and prior settings.

Forgiveness: reengagement and repaired confidence can dissociate The forgiveness paradigm adds a third phase: a cooperative partner becomes hostile and then returns to cooperation. This distinguishes two outcomes that ordinary payoff summaries blur. Reengagement is the willingness to return to a partner; trust repair is the restoration of confidence once the partner has returned. High α should shorten reengagement latency because recent positive evidence is weighted strongly. Cautious priors should lower the ceiling of repaired confidence even after reengagement occurs.

The final run reports reengagement rate [*Exp C: reengagement rate by condition*], payoff recovery [*Exp C: payoff recovery by condition*], beta recovery trajectories [*Exp C: beta recovery trajectories*], and reengagement latency [*Exp C: reengagement latency*]. The predicted dissociation maps onto the empirical distinction between forgiveness and trust repair: an agent can re-approach a partner before confidence has structurally recovered [18].

Mixed-volatility discrimination The mixed-volatility environment is the most direct behavioural test of relational precision. Four partners coexist: a stationary cooperator, a stationary exploiter, an abrupt-switch partner, and a gradual-drift partner. The agent must maintain confidence toward the stable safe partner while reducing commitment toward partners whose behaviour becomes unreliable. This environment should punish global or poorly calibrated confidence because evidence from one relationship is not informative about every other relationship.

The final run reports the discrimination index [*Exp D: discrimination index by condition*], concentration toward the stationary cooperator [*Exp D: concentration toward P0*], per-partner beta trajectories [*Exp D: per-partner beta trajectories*], and false-positive confidence reduction toward stable partners [*Exp D: false-positive rate*]. The predicted tradeoff is a social version of sensitivity versus specificity: high gain detects real change quickly but risks false positives; low gain preserves stability but misses genuine volatility; calibrated gain should best separate stable from changing partners.

From simulated phenotypes to empirical targets Together, the phenotype experiments turn the mechanism into a set of empirical targets. Individual differences in social learning rates and volatility tracking are already documented in human social learning [3, 14]. Computational psychiatry has also linked maladaptive social behaviour to aberrant precision weighting [1]. The present model adds a relationship-specific action-confidence layer: estimated α and prior parameters should predict not just learning speed, but patterns of approach, withdrawal, forgiveness, and false-positive social avoidance.

This is why the older mechanism experiments and the new phenotype experiments belong in the same paper. The mechanism experiments establish what

β_k does: it changes action confidence, carries cleaner local signal quality than global beta, and helps or harms depending on temporal structure. The phenotype experiments ask what kinds of agents result when the same mechanism is parameterised differently. The bridge between them is the core claim of the paper: affect is social metacognition, a relationship-specific regulator of when social beliefs become social commitments.

4 Discussion

Affect as social metacognition. Partner-local affective precision is best understood as a form of social metacognition: the agent’s estimate not of what a partner will do, but of how well its current model of that partner justifies confident action. The signal is turned inward, toward the agent’s own inference quality, rather than outward toward the partner’s mental states. This distinguishes the mechanism from theory-of-mind inference, which models what partners believe, know, or intend—a second-order model of another agent’s mental states [10, 16]. Affective precision does not do this. It tracks the focal agent’s confidence in its own model of a partner’s *behaviour*, not representations of that partner’s mental states. The signal is metacognitive rather than mentalistic.

The two mechanisms are complementary. An agent with sophisticated ToM inference could still be poorly calibrated about when to act confidently on those inferences, particularly if the partner is reliable in some contexts and unreliable in others. Partner-local affective precision addresses exactly this gap: it allows action confidence to track the recent track record of each partner model, independently of the model’s representational content [17]. The combination would permit an agent to both model partner intentions and maintain a running estimate of how much those models currently justify decisive action—a richer form of social metacognition that may be necessary for human-like performance in complex multi-partner environments.

The evidence supports this account as a mechanism rather than as a single reward-improvement effect. Across the current experiments, partner-local precision reliably changes deployment, the tracked-only lesion localises the effect to the $\beta_k \rightarrow \gamma_k$ pathway, and abrupt betrayal shows that deployment can improve payoff even when joint state accuracy falls. At the same time, the reliability-versus-reward diagnostic shows that the strongest model-fitness claim remains experimentally demanding: it requires tasks that separate action predictability from payoff exposure. The paper’s central contribution is therefore a structured account of social confidence dynamics, including the conditions under which confidence is useful, misleading, or merely diagnostically cleaner than a global alternative.

Individual differences and the computational basis of social trust variation. The phenotype taxonomy in Section 3.7 connects affective precision dynamics to a broader literature on individual differences in social cognition. The key observation is that variation in α and prior initial conditions does not produce

merely quantitative differences in outcomes—it produces qualitatively distinct behavioural signatures that are recognisable as human social types.

The anxious-reactive profile (high α , naive prior) parallels the computational signature of anxious hypervigilance in social contexts, in which threat-sensitive prediction error weighting produces rapid, high-variance oscillations in social confidence [1]. The hypervigilant profile (high α , cautious prior) parallels paranoid ideation—chronic wariness that resists updating even under sustained positive evidence. Computational psychiatry accounts of psychosis and social paranoia have proposed exactly this mechanism: aberrant precision weighting in the action-confidence pathway produces behaviour that prioritises threat detection at the cost of evidence integration. The avoidant-rigid profile (low α , cautious prior) corresponds to avoidant attachment [5]—not hypersensitive to negative signals but chronically unable to let positive evidence accumulate into confident engagement, producing social withdrawal maintained not by fear but by slow updating. The naive-stubborn profile captures a different failure mode: not paranoid but exploitable, failing to sufficiently update social confidence after negative evidence because the updating mechanism itself is sluggish.

Crucially, these phenotypes are not simply high-trust versus low-trust distinctions. They differ in the *asymmetry* and *speed* of trust revision, not just its baseline level. The forgiveness paradigm makes this particularly clear: α and prior produce independent effects on recovery dynamics that correspond to the empirical distinction between forgiveness (willingness to re-engage) and trust repair (restoration of confidence) [18]. High- α agents forgive quickly in the sense of returning to a partner; cautious-prior agents fail to repair trust fully even after returning. This dissociation is lost in any framework that treats social trust as a scalar quantity.

The mixed-volatility experiment adds a further dimension: individual differences in social confidence calibration translate into a sensitivity-specificity tradeoff with real consequences in social environments that contain both stable and changing relationships simultaneously. High- α agents are predicted to detect genuine partner shifts faster but to misattribute noise as signal at higher rates. This is the behavioural signature of social anxiety in trait personality research—enhanced vigilance to threat-relevant social cues at the cost of stable engagement with reliable partners [3]. The present framework positions these phenotypes as computational predictions for empirical individual-difference research: fitting α and prior parameters to individual learning curves in iterated trust games, then testing whether estimated parameters predict personality measures and clinical scores, would allow the model to be evaluated against human behavioural data.

The lesion result supports a deployment claim. The tracked-only lesion confirms that beta tracking alone is not the behavioural mechanism: removing the $\beta_k \rightarrow \gamma_k$ link makes tracked-only runs match no-affect controls, even though the beta state is still updated. The current evidence therefore supports a deployment interpretation: partner-local precision changes policy commitment through the action-confidence pathway. The lesion design is the methodological core of this claim—it rules out the alternative that β_k tracking indirectly biases be-

lief quality, which would have implicated the inference pathway rather than the commitment pathway.

The betrayal result reveals the temporal dependency. The corrected agent-choice selector reveals a specific temporal pattern in the abrupt betrayal task: local affect outperforms no-affect and tracked-only controls on payoff (1322.3 versus 1225.0) while simultaneously showing lower joint accuracy (0.319 versus 0.425). The payoff gain alongside the accuracy reduction confirms that the mechanism is not working through better belief updating. Instead, partner-local precision changes which commitments are made after social change. The affected agent reencounters the betraying partner less often than no-affect and tracked-only controls, indicating selective reallocation rather than global caution. This is exactly the behavioural role the mechanism was designed to make possible: confidence can be updated relationship by relationship, so a change in one partnership need not flatten action confidence across the entire social portfolio.

The implication is structural: any confidence mechanism that integrates evidence without a separate volatility channel will exhibit this temporal dependency. The β_k tracker’s failure mode under abrupt social change is not an implementation bug but an inherent property of social confidence built on evidence accumulation without change detection. This is structurally similar to the challenge faced by hierarchical volatility models [14, 3], but applied to action confidence rather than learning rates—a level of the hierarchy that existing models do not separately address.

Limitations. Four design features bound the current claims. First, β_k is an external auxiliary tracker; in the full Hesp et al. hierarchy it is a proper inferred hidden state, allowing the agent to form prior expectations over its own future affective states and to plan around anticipated changes in partner reliability. Making β_k a variational variable would enable richer prospective regulation of action confidence. Second, the current implementation keeps $q(\beta_k)$ categorical rather than continuous. This preserves the inverse-beta precision convention and keeps the update identifiable, but it discretises the precision dynamics. Third, partners are scripted rather than themselves being active-inference agents; the mechanism is tested in isolation from the reciprocal dynamics that characterise real social interaction. Fourth, the phenotype taxonomy in Section 3.7 is currently descriptive: we observe that different α and prior values produce qualitatively distinct behavioural profiles, and we connect these descriptively to human individual differences in social trust. A stronger claim—that individual differences in human social trust calibration arise specifically from variation in an affective precision mechanism with this structure—would require fitting the model to human behavioural data and establishing that parameter estimates predict theoretically relevant outcomes. The phenotype predictions are offered as a scaffold for such empirical work, not as an established correspondence.

Future directions. The most important near-term extension is empirical: fitting the affective precision model to human behavioural data from iterated trust

game paradigms with individual difference measures. Sections 3.7–3.7 generate specific, quantitative predictions about exploitation sensitivity, betrayal recovery time, forgiveness speed, and mixed-volatility discrimination that differ across phenotype quadrants and are measurable from behavioural time series without neuroimaging. The α and prior parameters could be estimated per participant and tested against personality measures (Big Five dimensions, attachment style) and clinical scores (social anxiety, paranoid ideation) to assess whether the computational phenotypes correspond to real psychological variation. Such a study would also allow a within-design test of whether global versus partner-local precision provides better fits to individual behavioural trajectories.

A second priority is the global- β ablation: whether partner-local precision is necessary, or whether a single shared precision estimate updated from aggregate social experience produces comparable stable-regime gains. The mixed-volatility environment in Section 3.7 constitutes a direct test of this question. The predicted outcome is that a shared tracker would over-suppress confidence toward reliably predictable partners or under-suppress it toward uncertain ones, depending on social environment composition, and crucially that it would fail to produce the discrimination pattern observed in the heterogeneous-volatility condition.

Third, the mixed-volatility environment points toward richer ecological settings: social environments with communication, reputation signalling, and coalition structure would introduce new sources of reliability evidence that the affective signal could in principle track. Extending partner-local β_k dynamics to incorporate evidence from reputation signals and third-party observations is a natural next step that connects to the indirect-reciprocity literature.

Finally, combining partner-local affective precision with ToM-style belief inference [16] would allow future work to study whether metacognitive action-confidence signals and mentalistic partner-state models interact during social decision-making. An agent with both mechanisms could be confident in its ToM inferences for some partners and uncertain for others—a richer form of social metacognition that may be necessary for human-like performance in complex multi-partner environments. Whether metacognitive confidence and mentalistic inference are functionally coupled or independent in human social cognition is an open empirical question that the present framework directly motivates.

5 Conclusion

Predicting what a partner will do and deciding how confidently to act on that prediction are separable problems. Partner-local affective precision addresses the second. By giving each social relationship its own model-fitness estimate, the mechanism allows action confidence to be calibrated independently across a social portfolio—more decisive toward partners whose behaviour the model predicts well, more cautious toward those who remain uncertain. This produces psychologically recognisable dynamics: reliability tracking, metacognitive rather than epistemic gains, engagement reorganisation before payoff signals, and principled betrayal failure that reveals the temporal assumption on which social

confidence depends. The parameters governing these dynamics—precision gain and prior over model fitness—produce a family of behavioural phenotypes that map onto individual differences in human social trust calibration, positioning the framework as a computational account of social affect that is both mechanistically grounded and empirically tractable.

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